5	**(a)**	**(i)**	Length of path 1 = 1.000 - 0.300 = 0.700 m Length of path 2 = 1.000 + 0.300 = 1.300 m Path difference = 1.300 – 0.700 = 0.600 m	B1
		(ii)	Phase difference $= \left(\frac{0.600}{1.000} \times 2\pi\right) \pm \pi$	M1
			$= 2.2\pi$ or 6.91 rad or 0.2π or 0.628 rad.	A1
		(iii)	Two segments / loops	M1
			One solid line + one dotted line	A1
		(iv)	Antiphase or π rad / 3.14 rad / 180° out of phase	A1
	(b)	**(i)**	For stationary waves to form, $L = 1.0 = n\left(\frac{\lambda}{2}\right)$ or $\lambda = \frac{2L}{n}$	M1
			From $v = f\lambda$	M1
			$f = \frac{v}{\lambda} = \left(\frac{n}{2L}\right)\sqrt{\frac{mg}{\mu}} = \left(\frac{n}{2}\right)\sqrt{\frac{mg}{\mu}}$ since $L = 1.000$	A0
		(ii)	$m = \frac{4f^2}{n^2}\left(\frac{\mu}{g}\right) = \frac{4(25)^2}{6^2}\left(\frac{7.0\times10^{-3}}{9.81}\right) = 0.050$ kg	A1
		(iii)	Next higher frequency has 8 segments	C1
			Frequency = 33 Hz	A1
6	**(a)**	**(i)**	From $V = IR$, $I = \frac{V}{}$ hence $I \propto \frac{1}{}$	C1